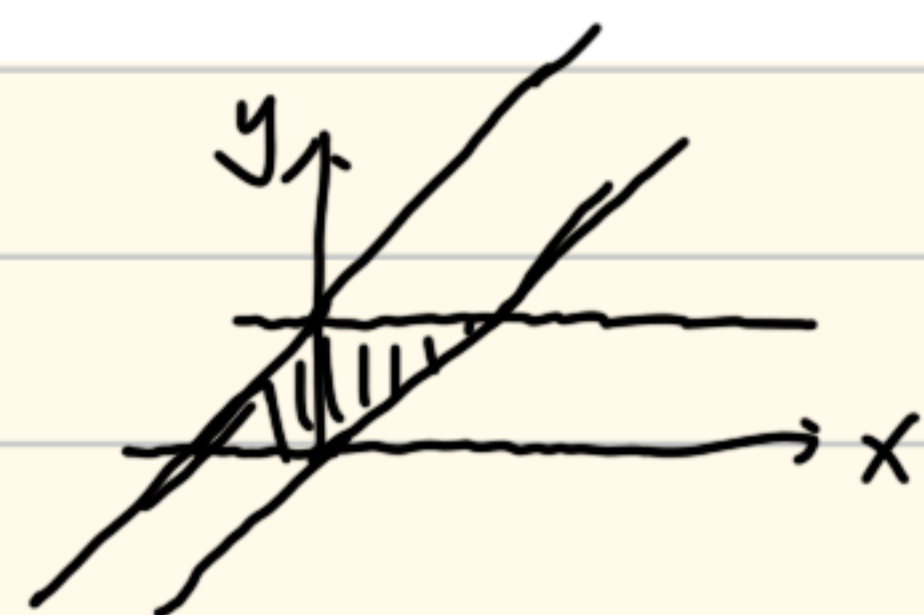
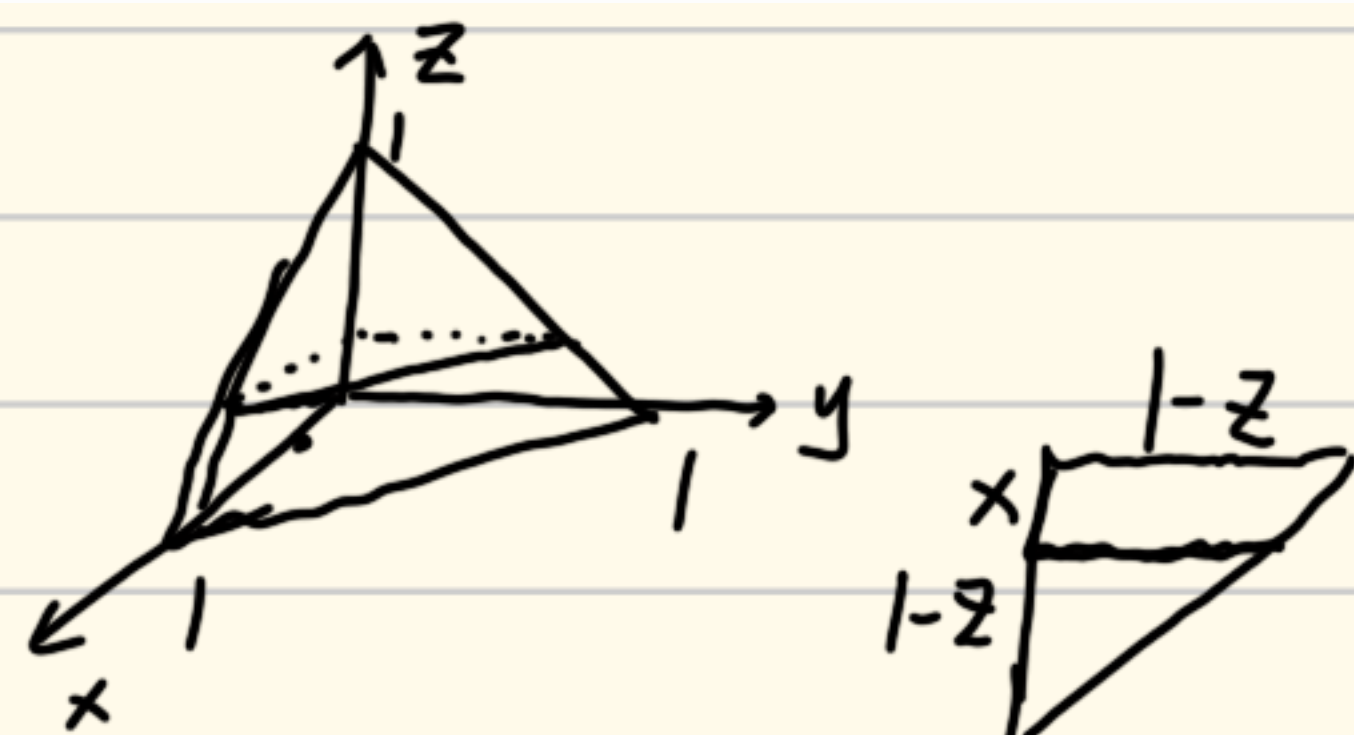


1.(10分) 设 D 是由直线 $y = 0, y = 1, y = x, y = x + 1$ 所围成的有界闭区域。求二重积分 $\iint_D (4y - 2x) dx dy$.



$$\begin{aligned}
 & \int_0^1 dy \cdot \int_{y-1}^y (4y - 2x) dx \\
 &= \int_0^1 (4y - y^2 + (y-1)^2) dy \\
 &= \int_0^1 (4y - y^2 + y^2 - 2y + 1) dy \\
 &= \int_0^1 (2y + 1) dy \\
 &= (y^2 + y) \Big|_0^1 \\
 &= 2
 \end{aligned}$$

2.(10分) 设 V 是由平面 $x = 0, y = 0, z = 0, x + y + z = 1$ 所围成的四面体。求三重积分 $\iiint_V \frac{1}{(1+x+y+z)^2} dx dy dz$.



$$\begin{aligned}
 \frac{\lambda}{1} &= \frac{1-z}{1} \\
 \lambda &= 1-z
 \end{aligned}$$

$$\begin{aligned}
 & \int_0^1 dz \int_0^{1-z} dx \int_0^{1-z-x} \frac{1}{(1+x+y+z)^2} dy \\
 &= -\int_0^1 dz \int_0^{1-z} \frac{1}{(1+x+y+z)} \Big|_0^{1-z-x} dx \\
 &= -\int_0^1 dz \int_0^{1-z} \left(\frac{1}{2} - \frac{1}{1+x+z} \right) dx \\
 &= -\int_0^1 \left(\frac{1}{2}x - \ln|x+1+z| \Big|_0^{1-z} \right) dz \\
 &= -\int_0^1 \left(\frac{1}{2}(1-z) - \ln 2 + \ln|z+1| \right) dz \\
 &= \int_0^1 \left(\frac{1}{2}(z-1) + \ln 2 - \ln(z+1) \right) dz \\
 &= \left(\frac{1}{4}z^2 - \frac{1}{2}z + z\ln 2 - (z+1)\ln(z+1) + z \right) \Big|_0^1
 \end{aligned}$$

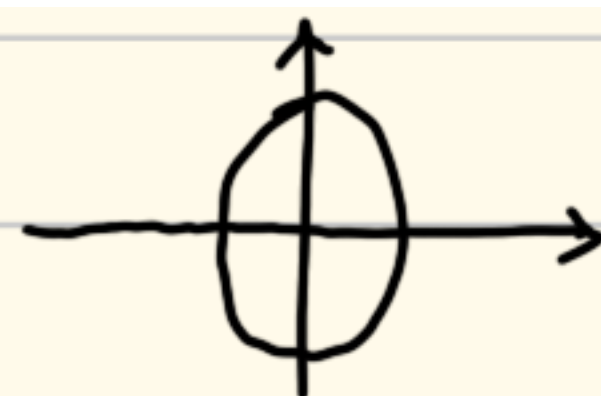
$$\begin{aligned}
 & \int \ln(z+1) dz \\
 &= z \ln(z+1) - \int z d\ln(z+1) \\
 &= z \ln(z+1) - \int \frac{z}{z+1} dz \\
 &= z \ln(z+1) - \int \left(1 - \frac{1}{z+1} \right) dz \\
 &= z \ln(z+1) - z + \ln(z+1)
 \end{aligned}$$

$$= \frac{1}{4} - \frac{1}{2} + \ln 2 - 2 \ln 2 + 1$$

$$= \frac{3}{4} - \ln 2$$

3.(10分) 设 E 是椭圆 $\{(x, y) \in \mathbb{R}^2 \mid x^2 + \frac{y^2}{4} = 1, -1 \leq x \leq 1, -2 \leq y \leq 2\}$. 求第一型曲线积分 $\int_E |xy| ds$.

3.



$$4 \int_E xy \, ds$$

$$x = \cos \theta, \quad y = 2 \sin \theta, \quad ds = \sqrt{\sin^2 \theta + 4 \cos^2 \theta} \, d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} 2 \sin \theta \cos \theta \cdot \sqrt{\sin^2 \theta + 4 \cos^2 \theta} \, d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} \sin 2\theta \sqrt{1 + 3 \cos^2 \theta} \, d\theta$$

$$= -\frac{4}{2} \int_0^{\frac{\pi}{2}} \sqrt{1 + \frac{3}{2}(1 + \cos 2\theta)} \, d \cos 2\theta$$

$$= -2 \int_0^{\frac{\pi}{2}} \sqrt{\frac{5}{2} + \frac{3}{2} \cos 2\theta} \, d \cos 2\theta$$

$$= -\frac{\sqrt{2}}{3} \int_0^{\frac{\pi}{2}} \sqrt{5 + 3 \cos 2\theta} \, d \cos 2\theta$$

$$= -\frac{\sqrt{2}}{3} \cdot \frac{2}{3} \left[(5 + 3 \cos 2\theta)^{\frac{3}{2}} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{2\sqrt{2}}{9} \cdot 7 \cdot 2^{\frac{3}{2}}$$

$$= \frac{14\sqrt{2}}{9} \cdot 2\sqrt{2} = \frac{56}{9}$$

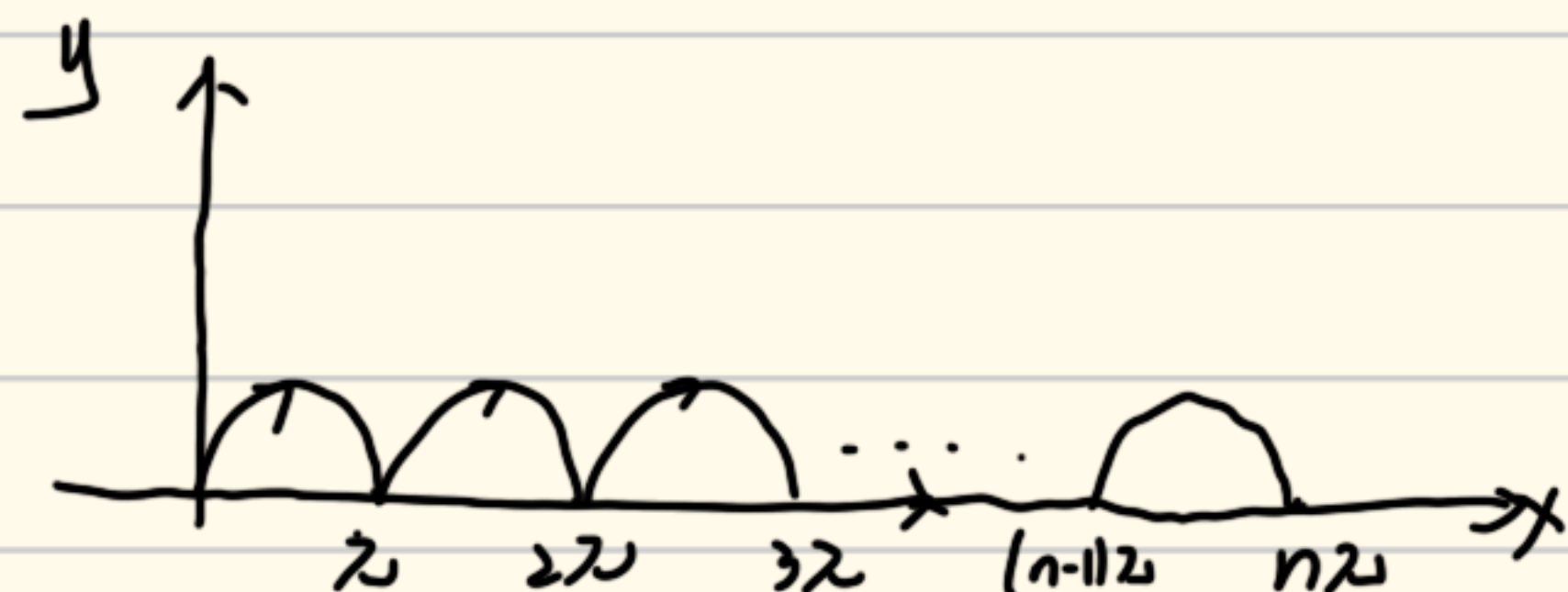
$$\sqrt{5 + \cancel{0}x}$$

$$2^{\frac{3}{2}} - 8^{\frac{3}{2}} = 2^{\frac{3}{2}}(1 - 4^{\frac{3}{2}})$$

$$= -7 \cdot 2^{\frac{3}{2}}$$

4.(15分) 设 n 是正整数, 从点 $(0, 0)$ 到点 $(n\pi, 0)$ 的有向曲线 $L_n = \{ (t, |\sin t|) \mid 0 \leq t \leq n\pi \}$. 计算出下面第二型曲线积分在 $n \rightarrow \infty$ 下的极限:

$$\lim_{n \rightarrow \infty} \int_{L_n} e^{y^2 - x^2} \cos(2xy) dx + e^{y^2 - x^2} \sin(2xy) dy$$



补充曲线 $y=0$, $0 \leq x \leq n\pi$, 顺时针方向, 记为 L_0^+

由格林公式

$$\int_{L_n + L_0^+} e^{y^2 - x^2} \cos(2xy) dx + e^{y^2 - x^2} \sin(2xy) dy$$

$$= \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) d\sigma$$

$$= \iint_S [-2x e^{y^2 - x^2} \cdot 2y \cos(2xy) + 2y e^{y^2 - x^2} \cdot 2x \sin(2xy)] d\sigma = 0$$

$$\therefore \int_{L_n} e^{y^2 - x^2} \cos(2xy) dx + e^{y^2 - x^2} \sin(2xy) dy$$

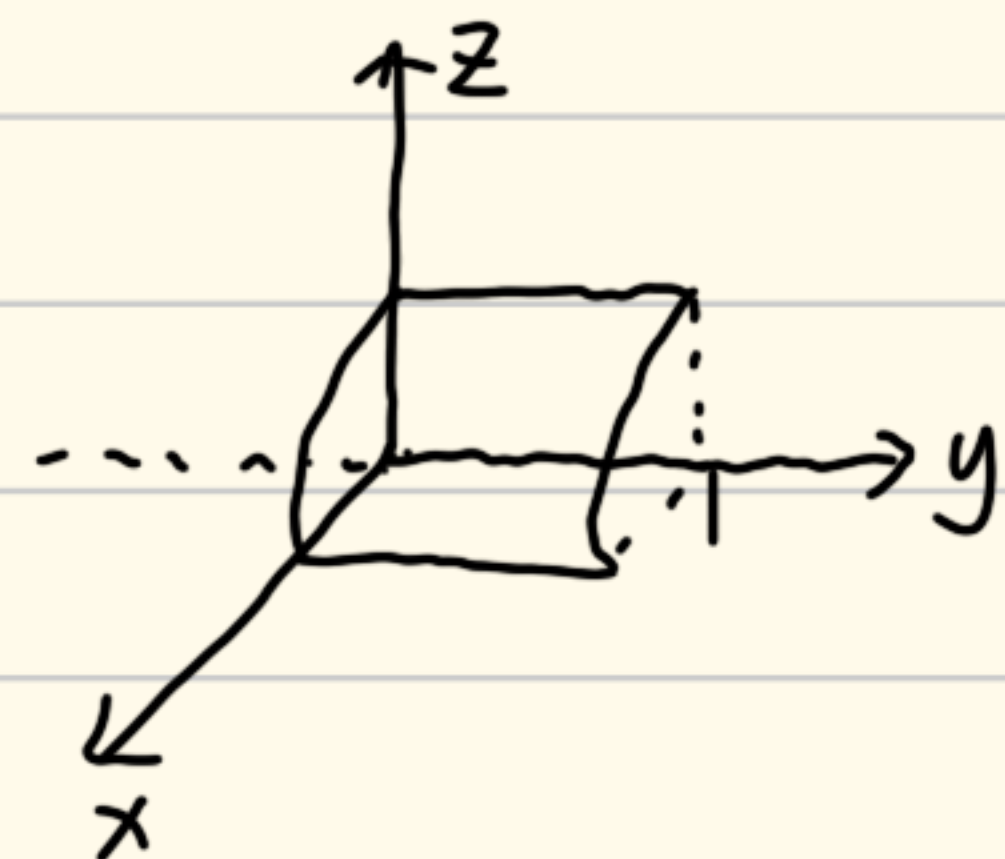
$$= \int_{L_0^+} e^{y^2 - x^2} \cos(2xy) dx + e^{y^2 - x^2} \sin(2xy) dy$$

$$= \int_{L_0^+} e^{-x^2} dx$$

$$= \int_0^{n\pi} e^{-x^2} dx$$

$$\lim_{n \rightarrow \infty} \int_0^{n\pi} e^{-x^2} dx = \int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \quad (\text{证明})$$

5.(10分) 设 S 是曲面 $\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + z^2 = 1, x \geq 0, z \geq 0, 0 \leq y \leq 1\}$. 求第一型曲面积分 $\iint_S x \, dS$.



$$z = \sqrt{1-x^2}, \quad dS = \sqrt{\frac{x^2}{z^2} + 1} \, dx \, dy = \sqrt{\frac{1}{z^2}} = \frac{1}{z} \, dx \, dy$$

$$2x + 2z x_z = 0, \quad z_x \cdot z = -x, \quad z_x = -\frac{x}{z}$$

$$\iint_S \frac{x}{\sqrt{1-x^2}} \, dx \, dy = \int_0^1 dy \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx = \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx = \frac{1}{2} \int_0^1 \frac{1}{\sqrt{1-x^2}} \, dx^2$$

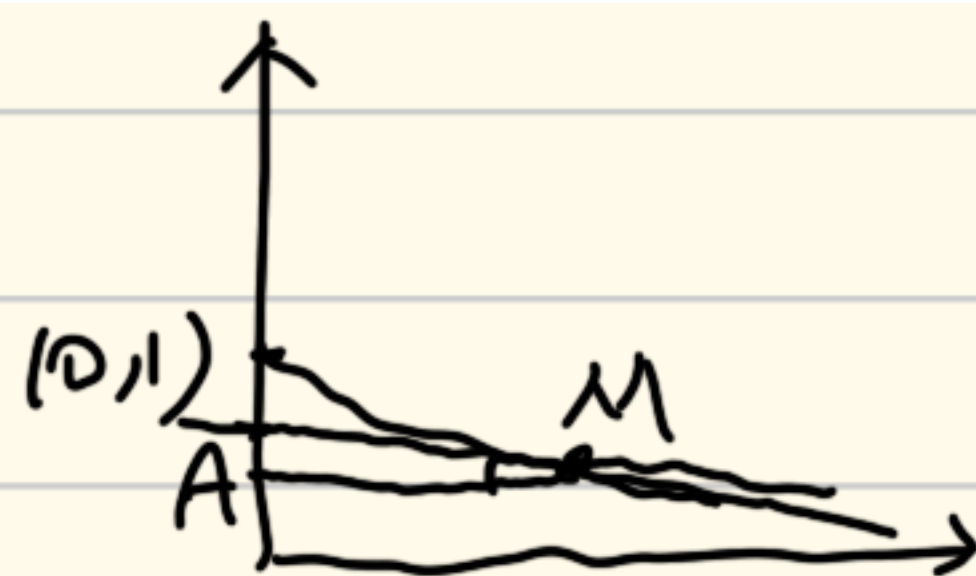
$$= -\frac{1}{2} \cdot 2 \cdot \sqrt{1-x^2} \Big|_0^1$$

$$= 1$$

6.(10分) 求第二型曲面积分 $\iint_{x^2+y^2+z^2=1 \text{ 外侧}} x \, dy \, dz + y \, dz \, dx + z \, dx \, dy$.

由高斯公式, 原式 = $\iiint_{x^2+y^2+z^2 \leq 1} 3 \, dV = 3 \cdot \frac{4}{3} \pi = 4\pi$

7.(15分) 假设平面直角坐标系第一象限中有一条曲线 $L = \{(x, y(x)) \mid x \geq 0\}$, 其中 $y(0) = 1$, $y(x)$ 是严格递减的、正的可导函数。任取 L 上一点 M , L 在 M 点的切线交 x 轴于点 A . 假定从 M 到 A 的直线段的长度恒为 1! 求出 $y = y(x)$ 所满足的一阶常微分方程, 并且解出这个方程的初值问题 $y(0) = 1$.



L 在 $M(x_0, y(x_0))$ 处的切线为

$$y - y(x_0) = y'(x_0)(x - x_0), \quad \text{令 } x = 0,$$

$$y(x_0) = y'(x_0)(x_0 - x)$$

$$x = x_0 - \frac{y(x_0)}{y'(x_0)},$$

$$\text{即 } A \cdot \left(x_0 - \frac{y(x_0)}{y'(x_0)}, 0\right) \quad M(x_0, y(x_0))$$

$$\therefore \text{有 } \sqrt{\left[\frac{y(x)}{y'(x)}\right]^2 + y^2(x)} = 1$$

$$\text{即 } \frac{y^2}{y'^2} + y^2 = 1, \quad y^2 + y^2 \cdot y'^2 = y'^2$$

$$\because y < 1, \text{ 则 } 1 - y^2 > 0, \quad y'^2 = \frac{y^2}{1 - y^2}, \quad \text{又有 } y' < 0,$$

$$\text{则 } y' = \frac{-y}{\sqrt{1 - y^2}},$$

$$\text{即有方程 } \frac{dy}{dx} = \frac{-y}{\sqrt{1 - y^2}}, \quad \text{又 } y(0) = 1$$

$$\frac{\sqrt{1 - y^2}}{y} dy = -dx,$$

$$\sqrt{1 - y^2} - \ln|\sqrt{1 - y^2} + y^2| + \ln y = -x + C$$

$$\therefore x + \sqrt{1 - y^2} - \ln|\sqrt{1 - y^2} + y^2| + \ln y = C$$

$$\text{代入 } x=0, y=1, \text{ 得 } C=0$$

$$\text{则有 } x = \ln|\sqrt{1 - y^2} + y^2| - \ln y - \sqrt{1 - y^2}$$

$$\int \frac{\sqrt{1 - y^2}}{y} dy = \int \frac{\cos^2 t}{\sin t} dt$$

$$= \int \frac{1 + \cos 2t}{2 \sin t} d(\sin t)$$

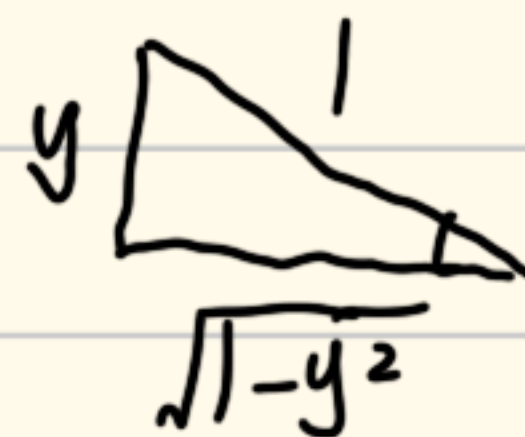
$$= \cos t - \int \sin t d \frac{\cos t}{\sin t}$$

$$= \cos t + \int \frac{1}{\sin t} dt$$

$$= \cos t - \ln |\cot t + \sin t|$$

$$y = \sin t, \quad t = \arcsin y$$

$$= \sqrt{1 - y^2} - \ln \left| \frac{\sqrt{1 - y^2}}{y} + y \right|$$



8.(10分) 求二阶常微分方程 $y'' + 4y = \sin 3x$ 的通解。

对应齐次方程为 $y'' + 4y = 0$, $\lambda^2 + 4\lambda = 0$, $\lambda_1 = 0, \lambda_2 = -4$

$\pm 2i$ 不是特征根

齐次方程通解为 $C_1 + C_2 e^{-4x}$

设 $y = A \cos 3x + B \sin 3x$,

$$y' = -3A \sin 3x + 3B \cos 3x$$

$$y'' = -9A \cos 3x - 9B \sin 3x$$

$$\therefore (-5A \cos 3x - 5B \sin 3x) = \sin 3x$$

$$\therefore \begin{cases} 5A = 0 \\ -5B = 1 \end{cases} \quad \begin{cases} A = 0 \\ B = -\frac{1}{5} \end{cases}$$

$\therefore$ 有特解 $y = -\frac{1}{5} \sin 3x$

$\therefore$ 通解为 $C_1 + C_2 e^{-4x} - \frac{1}{5} \sin 3x$

$$\lambda^2 + 4 = 0, \lambda_{1,2} = \pm 2i.$$

$$C_1 \cos \frac{2x}{2x} + C_2 \sin \frac{2x}{2x}$$

9.(10分) 两小题。

(1).(5分) 设 $D = \mathbb{R}^2 - \{(x, 0) \mid x \geq 0\}$. 写出一个函数 $T: D \rightarrow \mathbb{R}$ 满足 T 在 D 中每点可微, 并且

$$\frac{\partial T}{\partial x} = -\frac{y}{x^2 + y^2}, \quad \frac{\partial T}{\partial y} = \frac{x}{x^2 + y^2}.$$

$$T(x) = \frac{1}{2} \ln |x^2 + y^2| + C$$

(2).(5分) 设 $\Omega = \mathbb{R}^2 - \{0\}$. 证明不存在函数 $U: \Omega \rightarrow \mathbb{R}$ 满足 U 在 Ω 中每点可微, 并且

$$\frac{\partial U}{\partial x} = -\frac{y}{x^2 + y^2}, \quad \frac{\partial U}{\partial y} = \frac{x}{x^2 + y^2}.$$

在点 $(x, 0) \ x > 0$ 上, $\frac{\partial U}{\partial y} = 0, \frac{\partial U}{\partial x} = \frac{1}{x}$

$$U = C_1 y + \varphi(x) \quad \frac{\partial U}{\partial x} = \varphi'(x) = \frac{1}{x}, \quad \varphi(x) = \ln x + C$$

$$\int \frac{-y}{x^2+y^2} dy = -\frac{1}{2} \int \frac{1}{x^2+y^2} dy^2 = -\frac{1}{2} \ln |y^2+x^2| + C$$

$$\therefore U(x, y) = -\frac{1}{2} \ln |x^2+y^2| + \varphi(x)$$

$$\frac{\partial U}{\partial x} = -\frac{1}{2} \frac{2x}{x^2+y^2} + \varphi'(x) = \frac{x}{x^2+y^2}$$

$$\varphi'(x) = \frac{2x}{x^2+y^2}$$

$$\int \frac{1}{x^2+y^2} dx^2 = \ln |x^2+y^2|$$

$$\text{即 } U(x, y) = \frac{1}{2} \ln |x^2+y^2|$$